TASK 4: INVESTIGATION OF LIMITATIONS

1. Introduction

This section investigates the limitations of the Airbnb Listings and Reviews datasets used in the market intelligence analysis. Although the datasets provided useful information about advertised listing prices, accommodation characteristics, room types and guest reviews, several data-related limitations could affect the accuracy, reliability and interpretation of the findings.

The analysis used an Airbnb Listings dataset containing 2,953 records and a Reviews dataset containing 10,000 records. The following limitations were identified and evaluated.

2. Missing Listing Attributes

Some listing attributes contained missing or invalid values, including price, bedrooms, accommodation capacity and review ratings.

For example:

- Four listings had missing prices.
- Four listings had missing accommodation capacity.
- 669 listings had missing review ratings.
- All 2,949 listings with valid bedroom values had exactly one bedroom.

Impact on findings:

Missing information reduces the number of observations available for analysis and may introduce bias if listings with missing information differ from those with complete information.

The lack of variation in bedroom values also prevented the investigation of the relationship between the number of bedrooms and advertised prices.

3. Currency Differences

The listings dataset contained properties from several international cities, including Cape Town, Sydney, Paris, New York, Bangkok and Istanbul. However, the dataset did not clearly identify the currency used for every listing price.

Impact on findings:

Prices from different countries cannot be directly compared without confirming their currencies and converting them into a common currency using appropriate exchange rates.

For example, a listing price of 500 in Cape Town may not have the same monetary value as a listing price of 500 in Paris. Comparing these values without currency adjustment could produce misleading conclusions about which cities or room types are more expensive.

To reduce this limitation, the hypothesis tests comparing advertised prices were conducted within Cape Town rather than combining prices from different international markets.

4. Unequal Listing Counts Across Cities

The number of listings varied considerably between the cities included in the dataset.

City	Number of listings
Cape Town	513
Sydney	484
Mexico City	430
Istanbul	323
Rome	310
Rio de Janeiro	276
Bangkok	235
Paris	196
New York	122
Hong Kong	60
Total valid city listings	**2,949**

Impact on findings:

Cities with larger listing samples have more influence on overall descriptive results than cities with smaller samples.

For example, Cape Town has 513 listings, while Hong Kong has only 60. This means that findings for cities with smaller samples may be less stable and more sensitive to individual observations.

Unequal sample sizes may also affect the statistical power of comparisons between cities. Therefore, conclusions about differences between international markets should consider the number of available listings in each city.

5. Duplicate Review Identifiers

The Reviews dataset contained 10,000 review records and 10,000 unique review identifiers. No duplicate review identifiers were detected.

Impact on findings:

Duplicate review identifiers could cause the same review to be counted more than once, artificially increasing review activity and potentially distorting the analysis.

Although no duplicate review IDs were found, this does not guarantee that there are no other forms of duplication within the dataset. Checking the identifiers nevertheless improved confidence in the review activity calculations.

6. Different Levels of Review Activity

Review activity varied considerably across listings.

The review analysis identified:

- 7,084 unique listing identifiers.
- Approximately 68.86% of listings had only one recorded review.
- Approximately 23.14% had two recorded reviews.
- Only a small number had more than four reviews.

Furthermore, only 42 listing IDs matched between the Listings and Reviews datasets, representing approximately 0.59% of the unique listing IDs in the Reviews dataset. All 42 matching listings were from Paris.

Impact on findings:

Listings with more reviews may have greater representation in review-related analysis than listings with fewer reviews.

A listing with only one review may not provide enough information to understand its overall guest experience reliably.

The very low matching rate between the two datasets also limited the ability to investigate relationships between review activity and listing-level ratings across all cities.

Consequently, review activity should not be treated as a complete representation of Airbnb popularity or guest satisfaction.

7. Missing District Information

The dataset included a district field, but some listings had missing or unavailable district information.

Impact on findings:

Missing district information limits the ability to investigate differences in listing prices and accommodation characteristics across smaller geographical areas.

For example, properties in popular tourist districts may have different advertised prices from properties in residential areas.

Without complete district information, the analysis cannot accurately identify all neighbourhood-level pricing patterns. This also limits the usefulness of the findings for hosts who want to compare their properties with similar listings in their immediate area.

8. Potentially Influential Extreme Prices

The listing price analysis identified a wide range of advertised prices.

The overall price statistics included:

Measure	Price
Minimum	9
Median	229
Mean	420.61
Maximum	8,097

The mean was considerably higher than the median, suggesting a positively skewed price distribution.

Impact on findings:

Extremely high prices can increase the overall average and make a market appear more expensive than it typically is.

For example, a small number of luxury listings could increase the average price even when most listings are relatively affordable.

To reduce the influence of extreme values, the analysis considered median prices and used non-parametric statistical methods, including the Kruskal-Wallis H-test and Spearman correlation.

However, extreme prices were not automatically removed because some may represent genuine luxury accommodation rather than incorrect data.

9. Absence of Actual Booking and Occupancy Records

The Listings dataset provided advertised prices and property characteristics, but it did not contain actual booking transactions, occupancy rates or confirmed reservation revenue.

Impact on findings:

Advertised prices do not necessarily represent the prices guests actually pay.

A property may advertise a high nightly price but receive very few bookings. Another property may advertise a lower price but achieve higher occupancy.

Without booking and occupancy information, the analysis cannot determine:

- Which listings generate the highest revenue.
- Which cities have the highest occupancy rates.
- Whether expensive listings receive fewer bookings.
- Which room types are most commercially successful.
- Whether advertised prices reflect actual market demand.

Therefore, the research evaluates advertised listing prices rather than actual Airbnb financial performance or booking demand.

10. Overall Assessment of Limitations

The limitations identified above should be considered when interpreting the findings. Although the datasets provided useful insights into Airbnb listing characteristics, advertised prices and review activity, they do not provide a complete picture of actual market performance.

The analysis addressed some of these limitations by:

- Cleaning missing and invalid values where possible.
- Checking duplicate review identifiers.
- Using median prices alongside averages.
- Applying non-parametric statistical tests.
- Conducting price comparisons within Cape Town to reduce international currency differences.

However, limitations such as incomplete listing attributes, unequal city sample sizes, low matching between the Listings and Reviews datasets, missing district information and the absence of actual booking records could not be fully resolved.

Consequently, the findings should be interpreted as an analysis of the available Airbnb listing and review data rather than a complete representation of Airbnb market demand, revenue or profitability.

11. Conclusion

The investigation of limitations highlighted several factors that could influence the reliability and generalisability of the Airbnb market intelligence findings.

Despite these limitations, the datasets provided sufficient information to investigate advertised price differences, accommodation characteristics and review patterns. The use of appropriate data-cleaning procedures and statistical methods helped reduce some of the potential effects of these limitations.

Future research could improve the analysis by using datasets with verified currency information, more complete listing attributes, better matching between listings and reviews, and actual booking and occupancy records. These additional data sources would provide a more comprehensive understanding of Airbnb market performance.